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The daily rise and fall of the VIX1D: Causes and solutions of its overnight bias

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ABSTRACT

This paper explores the unique intraday dynamics of the VIX1D. We identify a distinct overnight bias, that causes the index to consistently rise during trading hours and to fall overnight. This bias stems from the index's calculation methodology, particularly the use of business time and dynamic weighting of next-term options, which include overnight variance risk premiums. It overlaps with and is more pronounced than the day-of-the-week effect. To mitigate this bias, we propose data filtering and revising the calculation method to a forward-starting variance. These solutions aim to enhance the VIX1D's interpretability and reliability for risk assessment in financial markets.

1. Introduction

Volatility indices are a popular tool in both academia and practice for obtaining direct assessments of future market risk - even in real-time. As they represent both the expected fluctuation and the risk premium contained in the underlying option prices, they are used for volatility forecasts, as a trading signal, for risk measures, for sentiment analyses, and as an underlying for derivatives (see Whaley (2000), Carr and Wu (2009), Obayashi et al. (2017), and Fassas and Siriopoulos (2021) among others). The most popular of these indices is the Cboe VIX Index. It is based on SPX options and reflects the implied volatility of the next 30 days of the S&P 500. Enabled by daily available zero day to expiration (0DTE) SPX options, the Cboe introduced the VIX1D in 2023, which, as a 1-day volatility index, expands the existing range of VIX indices (from nine days up to one year). By capturing this particularly important 1-day horizon, the VIX1D has the potential to significantly influence how scientists and practitioners generate volatility forecasts for the US equity market. The studies by Albers (2023) and Zhang et al. (2023) already highlight the advantages of VIX1D over the popular VIX and traditional time series models like the popular HAR model (Corsi, 2009). However, there is also significant criticism of the calculation method of VIX1D. In addition to the comparatively strong fluctuation of the index value, the extraordinary intraday pattern of the VIX1D is particularly discussed in internet forums and blogs. Within a day, the new index seems to continuously rise and then fall significantly overnight. This zig-zag pattern sometimes leads to misunderstandings in interpretation and questions the usefulness of the index.

Against this background, we analyse the characteristics and consistency of the VIX1D intraday pattern (Section 2), explain the overnight bias causing it (Section 3), and propose solutions to mitigate the overnight bias through simple data filtering and an adjustment to the calculation method (Section 4). Because the VIX1D was introduced in April 2023 and the close values were reconstructed by the Cboe until May 2022, our dataset for the daily close-to-close log returns covers the period from May 13, 2022,

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to November 14, 2023 (covering 379 days), and for the intraday analyses, the period from April 24, 2023, to November 14, 2023 (covering 237 days).

2. The intraday pattern of the VIX1D

Since its introduction in April 2023, the VIX1D has risen on 95% of all days during regular trading hours (RTH) and fallen again overnight. While the longer-term volatility indices VIX9D, VIX, VIX3M, VIX6M, and VIX1Y have decreased by an average of 1% within the day during this period, the VIX1D has risen each day by 28% on average. This extraordinary intraday pattern is particularly evident in Fig. 1 and is in stark contrast to those of the longer-term indices of VIX9D to VIX1Y as can be seen in Table 1 and by comparing the Figs. A.6 to A.10 in the Appendix.

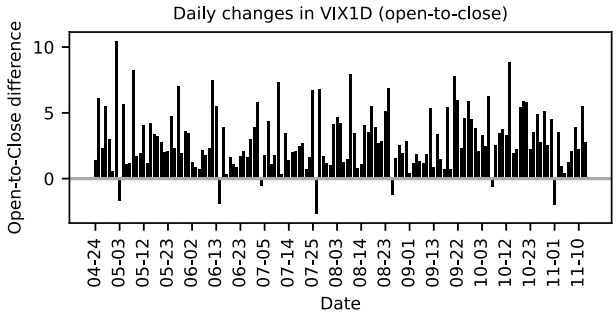

Fig. 1. Daily changes of the VIX1D (open-to-close) from 2023-04-24 to 2023-11-14.

Table 1
Daily changes of the six VIX indices from 2023-04-24 to 2023-11-14.

	Close-to-Open			Open-to-Close			Close-to-Close		
	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
VIX1D	−10.31	−2.98	3.86	−2.72	2.96	10.48	−9.76	−0.04	6.76
VIX9D	−4.47	0.51	4.20	−2.89	−0.48	2.58	−3.61	0.03	4.54
VIX	−3.36	0.23	2.99	−2.31	−0.22	2.79	−2.63	0.02	2.90
VIX3M	−3.01	0.11	2.38	−1.26	−0.08	2.53	−2.48	0.03	1.99
VIX6M	−2.25	0.11	1.95	−1.13	−0.08	1.85	−1.83	0.03	1.37
VIX1Y	−1.42	0.13	1.29	−0.93	−0.11	1.12	−1.18	0.02	1.15

The average intraday values of the volatility indices on a 5-minute basis provide a more detailed insight into this dynamic. As depicted in Fig. 2, the daily increase of the VIX1D follows a S-shaped, nearly linear pattern.

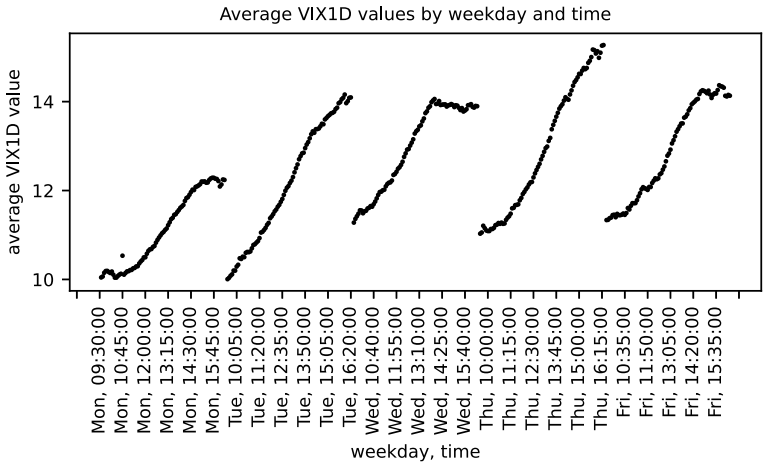

Fig. 2. Intraday pattern of the VIX1D from 2023-04-24 to 2023-11-14.

This distinct zig-zag pattern differs significantly from those observed in the longer-term VIX indices, which are illustrated in Figs. A.1 to A.5. While these indices tend to decrease over the course of the week and show a sharp increase over the weekend,

the VIX1D exhibits a zig-zag pattern rising from Monday to Friday and then significantly falling over the weekend. The daily close-to-close log returns show a day-of-the-week effect, which is particularly pronounced on Mondays and Fridays and increases in intensity with shorter time horizons of the indices. Fig. 3 illustrates this phenomenon, with the daily log returns of the VIX1D, VIX9D, and VIX significantly diverging from zero on Monday and Friday.

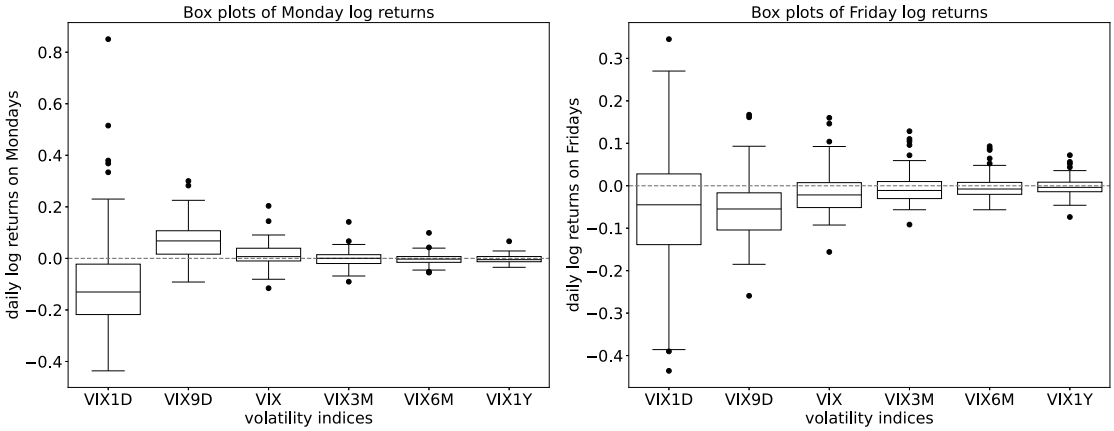

Fig. 3. Box plots of daily log returns on Monday and Friday of all volatility indices from 2022-05-13 to 2023-11-14.

Notably, there is a contrasting negative close-to-close return of the VIX1D over the weekend. For the longer-term indices with a time horizon of at least nine days, the day-of-the-week effect can mainly be attributed to the mismatch between the options' time to maturity calculated using the market convention (business days, i.e., 252 days) and Cboe's method for those indices (calendar days, i.e., 365 days). We refer to Gonzalez-Perez and Guerrero (2013), Kaplanski and Levy (2015), Fernandez-Perez et al. (2016), and Chen et al. (2021) for comprehensive analyses regarding the VIX. Since the focus of this study is on the more pronounced intraday pattern of the VIX1D, that is independent from the day-of-the-week effect, an in-depth analysis of the causes of the VIX1D's day-of-the-week effect is beyond the scope of this research.

3. Causes of the VIX1D overnight bias

The unusual intraday pattern of the VIX1D is neither based on a real increase in (expected) market risk nor on a special pricing of 0DTE options. The cause lies directly in the index's calculation methodology. Although this generally corresponds to that of the longer-term VIX indices, there are a few adjustments to account for the short time horizon of a single day (Cboe, 2023a). The adjustments with the greatest impact are the use of business (or trading) time instead of calendar time and the interpolation between the implied variance (IV) of near-term expiration options (that expire at the current day) and the IV of options with an expiry date closest to and after the near-term expiration (that expire at the next trading day).

The linear weighting of the IV's in relation to the trading time to expiry is described in Eq. (1) (Cboe, 2023a) and illustrated in Fig. 4.

$$VIX1D = 100 \times \sqrt{\left(T_1 \sigma_1^2 \left[\frac{M_{T_2} - M_{CM}}{M_{T_2} - M_{T_1}} \right] + T_2 \sigma_2^2 \left[\frac{M_{CM} - M_{T_1}}{M_{T_2} - M_{T_1}} \right] \right) \times \frac{M_{year}}{M_{CM}}}, \tag{1}$$

where M_X are the numbers of business minutes until expiration of the near-term options (M_{T_1}), until expiration of the next-term options (M_{T_2}), during RTH session (M_{CM}), and in one year (M_{year}). T_i is M_{T_i}/M_{year} and σ_i^2 are the near- and next-term variances of the corresponding SPX option contracts.

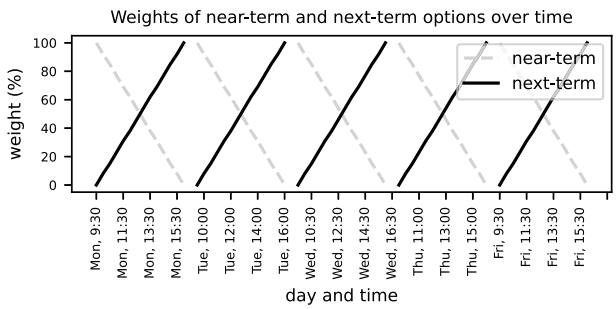

Fig. 4. Weights of near-term and next-term options over one week.

At the start of the RTH, the VIX1D is based only on the IV for the current day. Just as the time to expiration of near-term options decreases to zero during the day, so does their weight, while the proportion of the IV of the next-term options increases. At the end of the RTH, the VIX1D fully reflects the IV of the next-term options. This particular method of weighting based on the remaining time to expiration, combined with the employment of business time instead of calendar time, leads to the unusual intraday pattern. Because even though the IV of the next-term options is weighted based on the trading time to expiry, the option prices on which it is based contain a risk premium for potential market price fluctuations outside the RTH (i.e., overnight). Given that the next-term options incorporate overnight risk, unlike the near-term options, they tend to be more costly, reflected in their higher implied volatility. The fundamental trend of the VIX1D during the RTH (Fig. 2) thus primarily reflects the increasing weighting on these more expensive next-term options (Fig. 4). For further discussions on overnight market risks and the associated premium in option pricing, we refer to Andersen et al. (2015), Riedel and Wagner (2015), Lou et al. (2019), and Perras and Wagner (2020).

However, the intraday pattern caused by the overnight bias is not causing the overall higher fluctuation and extreme jumps in the closing values of the VIX1D. As described in Albers (2023) and visualized in Fig. A.11, the VIX1D is typically lower than the longer-term indices due to its shorter time horizon, but occasionally exhibits significantly larger jumps. The higher fluctuation and especially the extreme surges are largely due to the fact that market-relevant news have a more pronounced impact on the zero day and one day SPX options underlying the VIX1D than on the longer-dated SPX options used for the other indices (Cboe, 2023a)). This effect is likely intensified by the increasing proportion of 0DTE options that account for more than 40% of traded SPX options already (Cboe, 2023b). Since the availability of daily 0DTE SPX options provides pinpointed risk hedging for so-called known–unknowns, it is plausible that the VIX1D spikes on or shortly before the publication of key market data or decisions. For instance, the most extreme spikes in the period studied can be linked to the release of CPI data and FOMC press conferences.

Accordingly, the increased volatility of the daily values of the VIX1D can be reasoned by daily adjusted risk perceptions in a constantly changing market environment, while the consistent rise within trading days can be explained by the overnight bias.

4. Solutions to avoid the overnight bias

4.1. Using the same reference period

A straightforward approach to circumventing or significantly diminishing the overnight bias is to compare VIX1D values that are calculated at the same time each day, ensuring they reference the same period. Variations in these values can predominantly be attributed to the market participants' differing risk perceptions for those specific periods. Given that many scientific and practical applications (like volatility forecasting, Value-at-Risk predictions, and trading strategies) rely on daily values, the open and close prices of the VIX1D are of particular importance, as they relate solely to the current or upcoming trading day. For instance, Albers (2023) and Zhang et al. (2023) use only the closing values of the VIX1D for their volatility forecasts. They also demonstrate that the VIX1D overestimates the actual realized volatility of the S&P 500 due to the inherent variance risk premium. For analyses focusing solely on intraday volatility without potential overnight returns, the opening price of the VIX1D might be more suitable.

4.2. Changing the calculation method: VIX1D as 1-day forward-starting variance

Cboe's current method to calculate the VIX1D creates intraday drift due to the dynamic weighting on the next-term options that include the overnight variance risk premium. Consequently, comparing index values within the same day is not meaningful, and the observation that the VIX1D has increased over a single day does not provide significant insight. Therefore, we propose an adjustment to the index's calculation method. While such adjustments are rare, they have been made several times before. For example, in 2003, the Cboe changed the underlying for the VIX and included out-of-the-money options, in 2014 they also incorporated weekly options in addition to monthly options, and introduced smoothing algorithms for the VIX values (Cboe (2019), Cboe (2023c)). Instead of interpolating between the near-term and next-term options, we propose calculating the VIX1D as a 1-day *forward-starting* implied variance for the following trading day. The present VIX1D is based on a constantly rolling time frame: from reflecting the (expected) volatility σ_1 of the current day at open, to the (expected) volatility σ_2 of the next day at close, as formalized in Eq. (1). To avoid the bias caused by the overnight risk premium inherent in the next-term options and to enhance the information content of the VIX1D's intraday values, the revised VIX1D could be designed in such a way that it always reflects market participant's expectation of the close-to-close volatility of the next day.

The formula to approximate this is based on the principle that the variance over a period from t_0 to t_2 can be interpreted as the sum of the variance from t_0 to t_1 and the variance from t_1 to t_2. Hence, the relationship is:

$$(t_2 - t_0) \times \sigma^2_{0,t_2} = (t_1 - t_0) \times \sigma^2_{0,t_1} + (t_2 - t_1) \times \sigma^2_{t_1,t_2}.$$

Taking into account that the VIX1D is continuously calculated at time t_0 with current option prices, and using the notation of the Cboe from Eq. (1), it follows that:

$$\sigma^2_{\text{forward}} = \frac{\sigma_2^2 - w\,\sigma_1^2}{1-w},$$

and

$$VIX1D^{forward} = 100 \times \sqrt{\frac{\sigma_2^2 - w\,\sigma_1^2}{1-w}},$$

with $w = M_{T_1}/M_{T_2}$.

As this modified VIX1D, reflecting a *forward-starting* variance, consistently pertains to the same period during a single RTH session, the impact of the overnight risk premium embedded in next-term option prices on the VIX1D would no longer be increasing during the RTH. Consequently, the intraday prices of the VIX1D on the same day would be meaningfully comparable. An increase (decrease) in the index would indicate an actual rise (fall) in expected volatility, i.e., the anticipated risk. The disadvantage of using this methodology is that while the resultant VIX1D calculations will provide a consistent intraday measure of the next trading day's (expected) volatility, any information related to the current day's volatility will no longer be part of the resulting value. However, this potential drawback is of limited concern, as the previous day's closing value of the modified VIX1D could be utilized for the current day and adjusted for the overnight risk premium. Regarding these closing values, our proposed modification has the advantageous characteristic that the values of our alternative VIX1D align with those of the existing VIX1D, as 100% of the calculation weight is on the next day's implied volatility at those times.

5. Concluding remarks

The previous studies on the VIX1D by Albers (2023) and Zhang et al. (2023) have shown that the closing values of the new 1-day volatility index provide significant added value in forecasting the next day's volatility of the S&P 500, compared to existing volatility indices with longer time horizons and also compared to conventional time series models. The sometimes criticized higher variability and more pronounced spikes in the VIX1D's closing values, relative to the longer-term VIX indices, can be explained by the severe influence of market events on the ultra-short-dated SPX options underlying the VIX1D. This faster adaptation to the changing market conditions seems to be beneficial (see Albers (2023) and Zhang et al. (2023)).

However, our analysis of the intraday data of the VIX1D shows that the new 1-day volatility index exhibits a distinct intraday pattern and a significant day-of-the-week effect, both markedly different from the longer-term VIX indices. The defining feature is the intraday pattern, characterized by a consistent increase within and a notable decline outside of the RTH. We reveal that this daily increase is systematic and can be attributed to an overnight bias in the calculation methodology. This bias primarily stems from two causes: 1. The employment of business time, where each incremental unit of time is equally weighted regardless of whether RTH or overnight, and 2. The dynamic weighting on the typically more expensive next-term options, which incorporate an overnight variance risk premium. The resulting exceptional intraday pattern of the VIX1D often leads to misinterpretations. While intraday changes in the longer-term volatility indices are mainly due to changes in risk assessment by market participants (for example, in response to relevant market events), the intraday changes of the VIX1D are primarily caused by the overnight bias. Hence, the changes of the VIX1D within a day provide almost no new information, yet lead to confusion.

To minimize the distortion caused by the overnight bias, only daily values of the VIX1D calculated at the same time should be compared or employed in forecasting applications. Particularly, the opening and closing values are valuable, as they refer to the current and next trading day, respectively. This method of data restriction is very simple and suffices for most scientific research, but in the context of rapid information processing in financial markets, it forfeits the potential insights gained from continuous real-time index calculation and dissemination for practitioners.

Therefore, we suggest a simple modification of the VIX1D methodology to avoid the overnight bias. Instead of interpolating between near-term and next-term options, we propose calculating the VIX1D as a forward-starting implied variance. This modified index would consistently reflect the expected close-to-close volatility of the next trading day. Therefore, it has particularly two advantages over the current variant: 1. Since the overnight risk premium is uniformly included, the index is not distorted by the overnight bias. Intraday changes would be due to an actual change in expected volatility instead of systematically rising and would thus have relevant information content. 2. The (expected) close-to-close volatility of the next trading day, to which the modified VIX1D continuously refers, is much more relevant for numerous practical applications (such as Value-at-Risk forecasts, Expected Shortfall forecasts, and various trading strategies) than the rolling period referred to by the current VIX1D. While both indices refer to the next trading day at the end of the trading day, the adjusted index thus already provides a pinpointed gauge of the expected volatility of the next trading day before the markets close. The already high information content of the VIX1D could thus be further enhanced, offering both scientists and practitioners valuable insights into the (perceived) short-term market risk of the US equity market in real time.

While we provide clarity regarding the interpretation of the VIX1D's daily increases, highlight weaknesses in its current calculation method, and present concrete solutions to circumvent the overnight bias, applying this modification to the necessary extensive options dataset represents a promising approach for further research.

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We would like to thank the Editor-in-Chief, Samuel Vigne, an anonymous reviewer, and Henry Schwartz from Cboe for their constructive comments and suggestions.

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CRediT authorship contribution statement

Stefan Albers: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lars N. Kestner:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used are obtained from Refinitiv Eikon.

Appendix

See Figs. A.1–A.11.

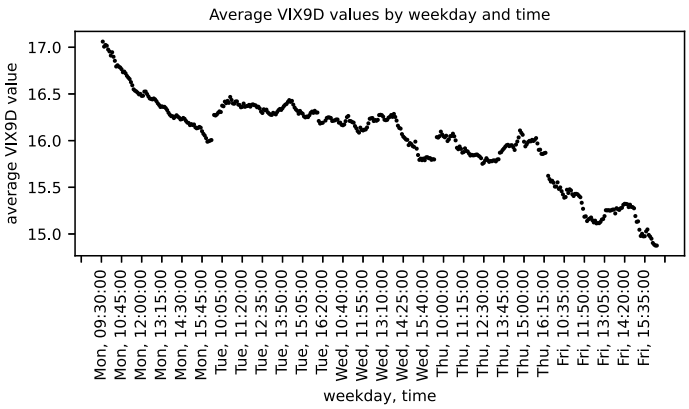

Fig. A.1. Intraday pattern of the VIX9D from 2023-04-24 to 2023-11-14.

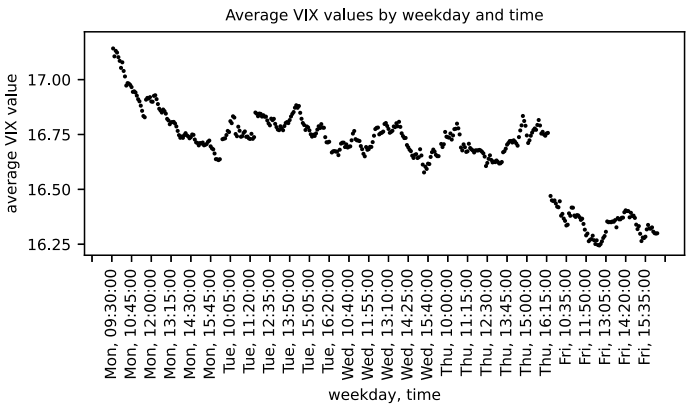

Fig. A.2. Intraday pattern of the VIX from 2023-04-24 to 2023-11-14.

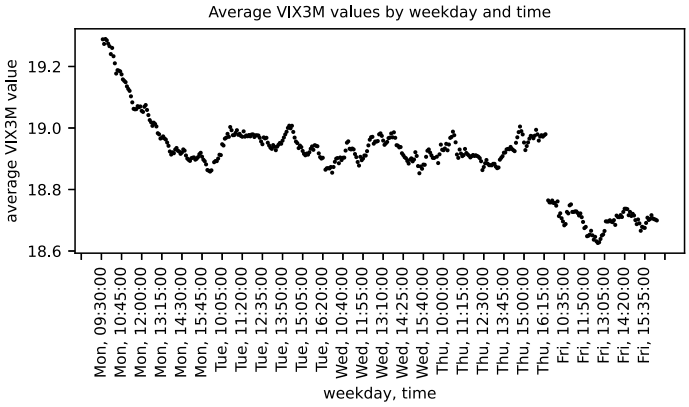

Fig. A.3. Intraday pattern of the VIX3M from 2023-04-24 to 2023-11-14.

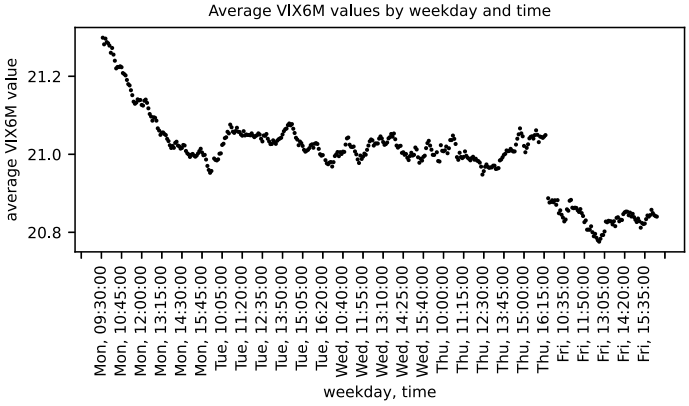

Fig. A.4. Intraday pattern of the VIX6M from 2023-04-24 to 2023-11-14.

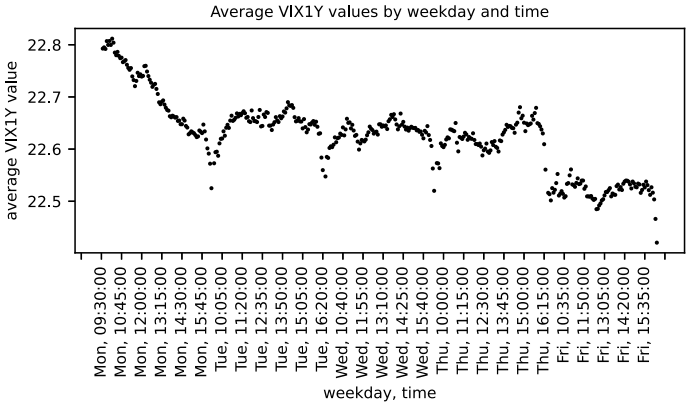

Fig. A.5. Intraday pattern of the VIX1Y from 2023-04-24 to 2023-11-14.

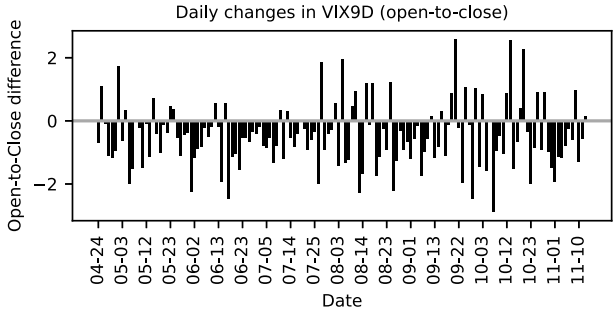

Fig. A.6. Daily changes of the VIX9D (open-to-close) from 2023-04-24 to 2023-11-14.

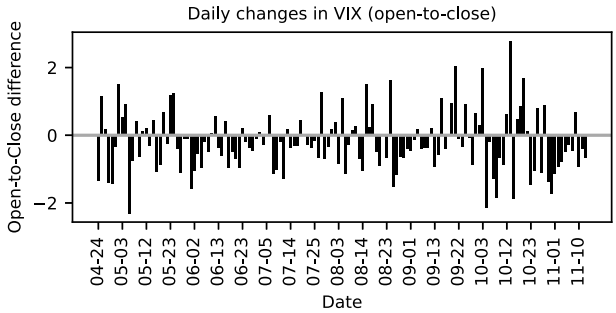

Fig. A.7. Daily changes of the VIX (open-to-close) from 2023-04-24 to 2023-11-14.

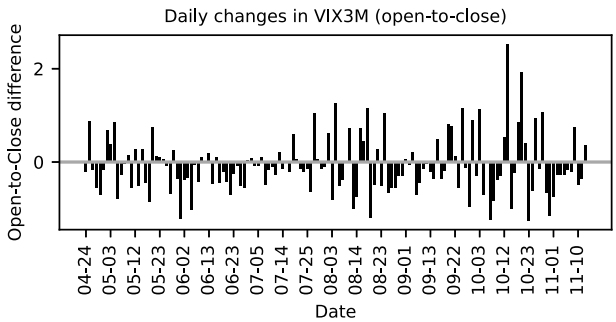

Fig. A.8. Daily changes of the VIX3M (open-to-close) from 2023-04-24 to 2023-11-14.

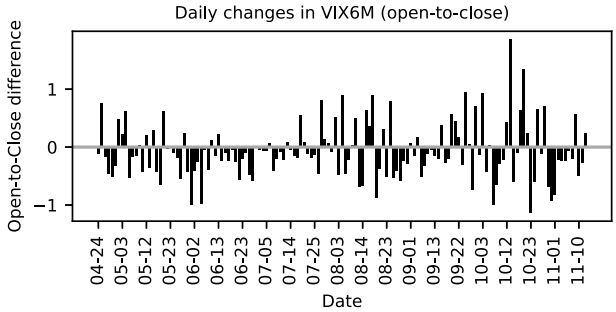

Fig. A.9. Daily changes of the VIX6M (open-to-close) from 2023-04-24 to 2023-11-14.

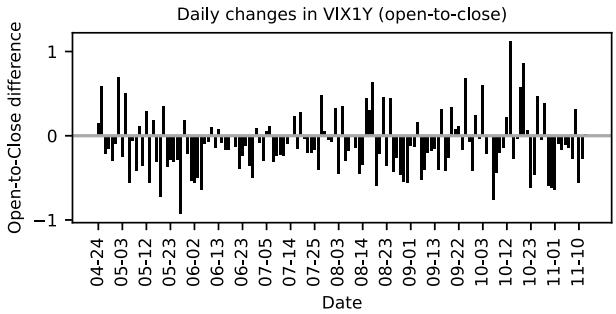

Fig. A.10. Daily changes of the VIX1Y (open-to-close) from 2023-04-24 to 2023-11-14.

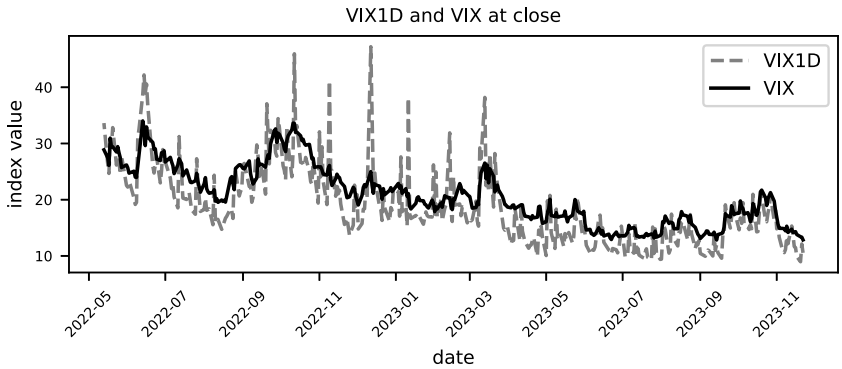

Fig. A.11. Daily close values of the VIX1D and VIX from 2022-05-13 to 2023-11-22.

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